PROJECT DESIGN PHASE

Proposed Solution Template

Date	27 September 2026
Team ID	23KD1A1507
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	2 Marks

Proposed Solution Template:
The project team has filled in the following information for the proposed solution.

S.No.	Parameter	Description
1	**Problem Statement** (Problem to be solved)	Traditional or partially manual network request management requires employees or IT staff to repeat tasks such as filling out physical forms, routing emails for approval, working out connection specifications, recording ticket details, and manually tracking request statuses. These manual steps can cause delays, inconsistent configurations, and difficulty tracking request records. A centralized, automated system is needed to manage requests end-to-end, enforce approval workflows consistently, maintain a unique identifier for every ticket, and update statuses automatically.
2	**Idea / Solution description**	A scoped ServiceNow application that manages network request workflows end-to-end: • **Service Catalog & Variables:** Network Request catalog item with dynamic variable sets. • **Custom Database Table (`u_database_table`):** Stores database number, requester, mobile number, connection type, customer address, payment mode, total amount, and approval status. • **Flow Designer Automation:** Triggers on catalog submission, fetches catalog variables, creates custom database records, routes approvals (manager/security), sends email notifications, and updates record status. • **Access Controls (ACLs):** Enforces secure role-based permissions across network tables.
3	**Novelty / Uniqueness**	The project applies a low-code enterprise platform (ServiceNow) to IT infrastructure service requests, so no backend infrastructure or database has to be built from scratch. Catalog items and database tables are configuration-driven rather than hardcoded, allowing new network categories and approval chains to be added without rebuilding the application. Request routing and notifications run automatically through Flow Designer event-driven workflows.

S.No.	Parameter	Description
4	**Social Impact / Customer Satisfaction**	Automation reduces repetitive manual effort for IT support staff and shortens the time needed to fulfill infrastructure requests. Requesters benefit from structured forms, transparent approval tracking, and unique ticket numbers that make each request easy to monitor. As a prototype, it demonstrates how digital workflow automation reduces IT bottlenecks and configuration errors; real-world enterprise impact is verified through system audit logs and execution history.
5	**Business Model (Revenue Model)**	The current project is an academic prototype and does not generate revenue. Possible models if developed further: a subscription or enterprise licence for organizations needing automated IT service management; professional consulting and implementation services for ITOM workflows; and custom integration connectors for external infrastructure tools.
6	**Scalability of the Solution**	New network services, catalog variables, and approval groups can be added as records without changing the application logic. ServiceNow provides the underlying cloud platform infrastructure for data storage, Flow Designer workflows, and email notifications. The same architecture can be adapted to hardware device provisioning, software license requests, and campus-wide IT services, and extended with external REST integrations. Production deployment requires standard security, performance, and load testing.